

JC4004 – Computational Intelligence

Sessions 3 & 4: Probability theory

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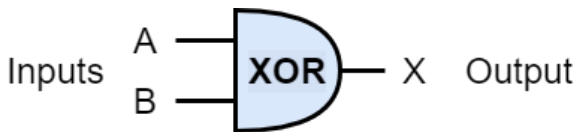
Probabilities in computational intelligence

- Many phenomena in the real world appear to be probabilistic
 - For example: variations in weather, mutations in biological organisms, even actions of humans on the population level
 - Even if the underlying mechanisms are deterministic, many such phenomena are easiest to analyse and understand by stochastic models
- Non-symbolic AI deals with probabilities
 - Even if the system output is presented in an explicit format, the internal mechanisms of the system often process data in a stochastic manner
 - Basic understanding of probability theory is essential for understanding many concepts and techniques in non-symbolic AI

Fundamentals of probabilities

- In deterministic systems, we can compute the outcome, if we know how the system works, and we know the inputs of the system

Example: logical gates



A	0	0	1	1
B	0	1	0	1
X	0	1	1	0

- In stochastic (random) systems, we cannot compute the outcome, but we can predict *probabilities* of different possible outcomes

Example: rolling a dice



Random variables

- A *random variable* (or *stochastic variable*) X is a variable whose value depends on the outcome of a stochastic event, like rolling a dice
- A *discrete* random variable with sample space S can be described by a *probability mass function* (pmf) μ :

$$\mu: S \rightarrow \mathbb{R}, \quad \mu(x) \geq 0 \quad \text{and} \quad \sum_{x \in S} \mu(x) = 1$$

- A *continuous* random variable and its distribution can be described by a *probability density function* (pdf) f :

$$f: S \rightarrow \mathbb{R}, \quad f(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} f(x) = 1$$

Cumulative distribution function

- Any random variable can be described by its *cumulative distribution function (cdf)* $F_X(x)$, giving the probability that the random variable is less than, or equal to, a certain value x :

$$F_X(x) = \Pr(X \leq x), \quad x \in \mathbb{R}$$

- For a discrete random variable, we can derive cdf from pmf as:

$$F_X(x) = \sum_{\omega \in \mathcal{S}; \omega \leq x} \mu_X(\omega)$$

- For a continuous random variable:

$$F_X(x) = \int_{-\infty}^x f_X(\omega) d\omega$$

Expectation

- *Expectation* $E(X)$ is a weighted average of the possible values of X , i.e., summation of each value weighted by its probability

$$E(X) = \sum_{x \in S} x \cdot \mu_X(x)$$

- If we repeat a stochastic experiment many times, the average outcome approaches the expectation
- For continuous variable X , summation is replaced by integration:

$$E(X) = \int_{-\infty}^{\infty} x \cdot f_X(x) dx$$

Variance and standard deviation

- Expectation only gives information about the average expected outcome: it does not indicate anything about the *shape* of the probability distribution
- *Variance* $\text{Var}(X)$ (or σ^2) is a measure of how widely the random outcomes are spread from the expectation:

$$\text{Var}(X) = E[(X - E(X))^2]$$

- *Standard deviation* σ is the average distance between the random outcomes and their expectation:

$$\sigma = \sqrt{\text{Var}(X)}$$

Approximating expectation and variance

- Often, we do not know the pdf of the random variable X , but we have some samples (measurements) representing X
 - Given samples $\mathbf{x} = \{x_1, \dots, x_n\}$, we can approximate the expectation by computing the average:

$$E(X) \approx \frac{1}{n} \sum_{i=1}^n x_i$$

- Respectively, we can approximate variance:
$$\text{Var}(X) \approx \frac{1}{n} \sum_{i=1}^n \left(x_i - \frac{1}{n} \sum_{j=1}^n x_j \right)^2$$
- The more samples we have, the more accurate the approximations are

Expectation and variance example

- Given the daily maximum temperatures measured for one week in Foshan in November, approximate the expectation and variance:

$$x = \{22, 21, 19, 22, 19, 21, 23\}$$

$$E(X) \approx \frac{22 + 21 + 19 + 22 + 18 + 21 + 23}{7} = \frac{147}{7} = \underline{\underline{21}}$$

$$\text{Var}(X) \approx \frac{(22 - 21)^2 + \dots + (23 - 21)^2}{7} = \frac{1 + 0 + 4 + 1 + 4 + 0 + 4}{7} = \frac{14}{7} = \underline{\underline{2}}$$

Common probability distributions

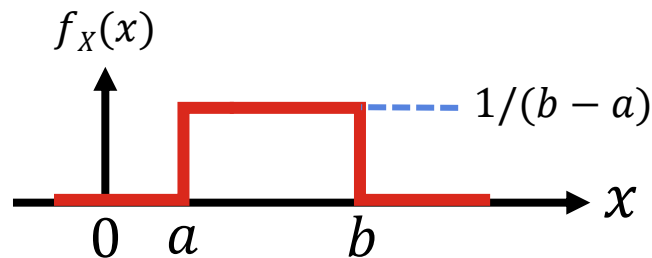
- There are several commonly used probability distributions with a pdf expressed in a closed form, characterised by some parameters
- Different probability distributions represent different kinds of stochastic events or phenomena
 - *Uniform distribution* represents equal probability for all the possible outcomes, given the range of possible values as a parameter
 - *Gaussian distribution* or *normal distribution* approximates distribution of many kinds of random variations occurring in nature
 - And many others, e.g., Bernoulli distribution, binomial distribution, Poisson distribution, exponential distribution, Gamma distribution...

Uniform distribution

- In *uniform distribution*, all possible outcomes are equally probable
 - For example, when rolling a dice or picking up a lottery ticket, all the outcomes among the n possible outcomes have equal probability $1/n$
 - For continuous random variables, we can define probability density function for uniform distribution in range $[a, b]$ as:

$$f_X(x) = 1/(b - a) \quad \forall a \leq x \leq b$$
$$f_X(x) = 0 \quad \forall x < a, x > b$$

- If X follows uniform distribution, we can use notation $X \sim \mathcal{U}(a, b)$

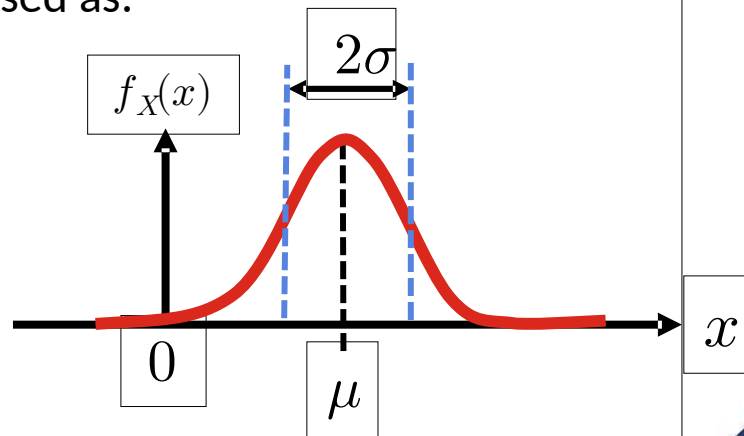


Gaussian distribution

- Practical experience has shown that *Gaussian (normal) distribution* can approximate natural variation very accurately
 - Gaussian distribution is parametrised with mean (expectation) μ and variance σ^2 , and its pdf can be expressed as:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- If X follows Gaussian distribution, we can use notation $X \sim \mathcal{N}(\mu, \sigma^2)$



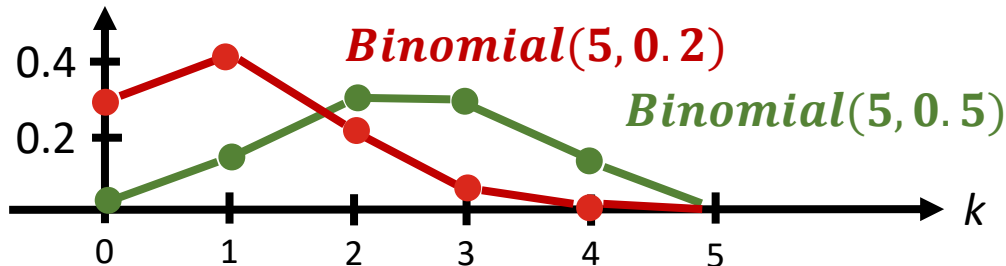
Bernoulli and binomial distribution

- *Bernoulli distribution* is a discrete distribution with only two possible outcomes, 0 and 1 (e.g., tossing a coin)

$$\Pr(0) = p \text{ and } \Pr(1) = q = 1 - p$$

- *Binomial distribution* defines the probability of k successes in a sequence of n Bernoulli experiments

- When $X \sim \text{Binomial}(n, p)$, $\Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$



Break: any questions?

Joint probability distributions

- Joint probability distribution is a probability distribution of two (or more) outputs (random variables) of a single random event
- Let us assume two random variables, X and Y , with a joint distribution, described by the joint pdf:

$$f_{XY}(x, y) = \Pr(X = x, Y = y)$$

- The random variables X and Y are said to be independent, if:

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

Conditional probability

- The conditional probability distribution $f_{Y|X}(y|x)$ gives the probability that $Y = y$, *given* that $X = x$

$$f_{Y|X}(y|x) = \Pr(Y = y|X = x)$$
$$f_{Y|X}(y|x) = f_{XY}(x, y)/f_X(x)$$

- Probability that events A and B both happen can be expressed as:

$$\Pr(A, B) = \Pr(A|B) \Pr(B)$$
$$\Pr(A, B) = \Pr(B|A) \Pr(A)$$

Conditional probability example

- What is the probability that a student passed the course, given that:

- A) They attended the lectures
- B) They did not attend the lectures

	attend	not attend	sum
passed	60	15	75
failed	5	20	25
sum	65	35	100

$$A) \Pr(\text{pass} \mid \text{attend}) = \frac{\Pr(\text{pass}, \text{attend})}{\Pr(\text{attend})} = \frac{60/100}{65/100} \approx \underline{\underline{92\%}}$$

$$B) \Pr(\text{pass} \mid \text{not attend}) = \frac{\Pr(\text{pass}, \text{not attend})}{\Pr(\text{not attend})} = \frac{15/100}{35/100} \approx \underline{\underline{43\%}}$$

Covariance

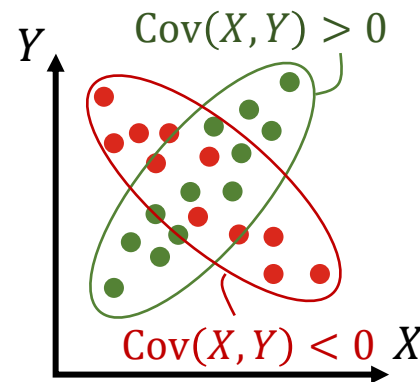
- *Covariance* $\text{Cov}(X, Y)$ is a measure of joint variability of two random variables X and Y :

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

- We can also simplify the definition as:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

- Large absolute value of the covariance indicates a linear relationship between X and Y
 - The relationship can be negative or positive, depending on the sign



Covariance

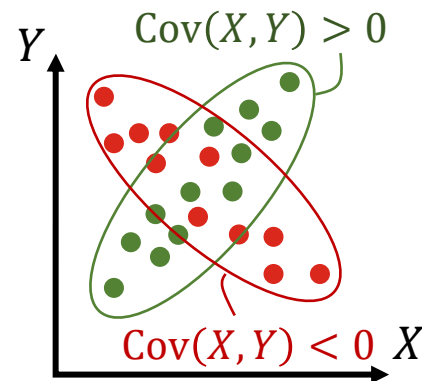
- Covariance of two random variables X and Y .
Note that $\text{Cov}(X, X) = E[(X - E[X])^2] = \text{Var}(X)$

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

- We can also simplify the definition as:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

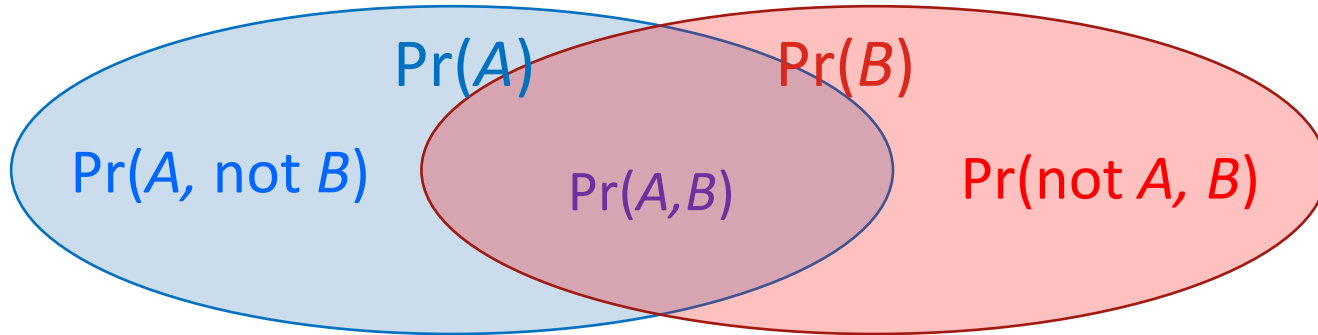
- Large absolute value of the covariance indicates a linear relationship between X and Y
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Bayes' theorem

- Given two random events, A and B , we can relate their conditional probabilities using Bayes' theorem:

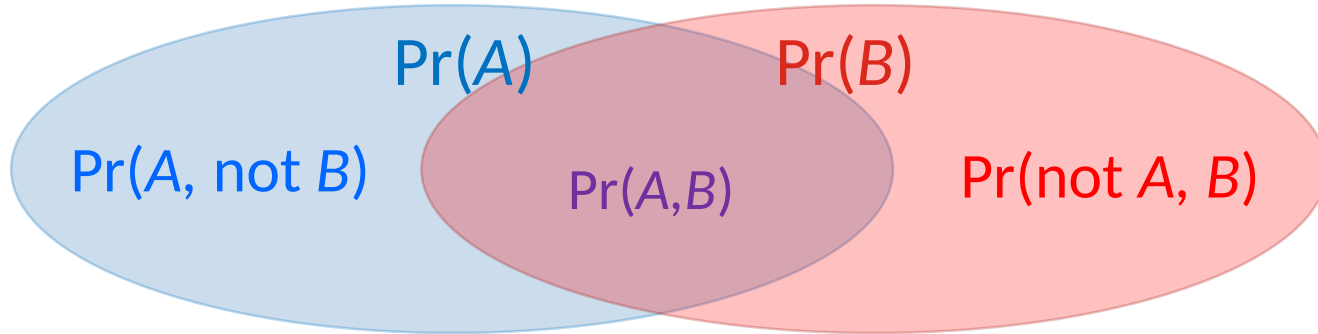
$$\Pr(A|B) = \frac{\Pr(B|A) \Pr(A)}{\Pr(B)}$$



Bayes' theorem

- Given two random events, A and B , we can relate their conditional probabilities using Bayes' theorem:

$$\Pr(A|B) = \frac{\Pr(B|A)\Pr(A)}{\Pr(B)} = \Pr(B|A)\Pr(A) + \Pr(B|\text{not } A)\Pr(\text{not } A)$$



Bayes' theorem example

- We have developed a deep learning method to detect cancer; unfortunately, the method is not fully accurate
 - The true positive rate (positive for people with cancer) is 90%
 - The false positive rate (positive for people without cancer) is 1%
 - From statistical studies, we know that 0.1% of population has cancer
- What is the probability that a person testing positive actually has cancer?

$$\begin{aligned}\Pr(\text{cancer} | \text{positive}) &= \frac{\Pr(\text{positive} | \text{cancer}) \Pr(\text{cancer})}{\Pr(\text{positive} | \text{cancer}) \Pr(\text{cancer}) + \Pr(\text{positive} | \text{no cancer}) \Pr(\text{no cancer})} \\ &= \frac{0.9 \cdot 0.001}{0.9 \cdot 0.001 + 0.01 \cdot 0.999} \approx 0.0826 \approx \underline{\underline{8.3\%}}\end{aligned}$$

Bayesian inference

- A common application of Bayes' theorem is *Bayesian inference*: probability of a hypothesis is updated using Bayes' theorem
 - Assuming observations x follow a hypothetical distribution $p(x|\theta)$, where θ is the parameter of the distribution, we can relate the distributions as:

$$p(\theta|x) = \frac{p(x|\theta) p(\theta)}{p(x)}$$

- We know x , but we do not know θ ; however, we may be able to guess or estimate the distribution $\theta \sim p(\theta|\alpha)$ where α is a *hyperparameter*
- We can update our estimate of $p(\theta|x)$ as we receive more observations x

Bayesian inference

- A common application of Bayes' theorem is *Bayesian inference*.
probability of a hypothesis $p(\theta)$ is the *prior*, i.e., the initial estimated probability that the model parameter is θ where θ is the parameter of the distribution, we can relate the distributions as:
- Assuming observation

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}$$

$p(\theta|x)$ is the *posterior*, i.e., the updated estimated probability that the model parameter is θ , after observing x

we may be able to guess or
is a *hyperparameter*
receive more observations x

Bayesian inference

- A common application of Bayes' theorem is *Bayesian inference*:
 $\Pr(x|\theta)$ is the *likelihood* of observing x , assuming that the model parameter is θ
Bayes' theorem
prior distribution $p(x|\theta)$, where
 θ is the parameter of the distribution, we can relate the distributions as:

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}$$

- We know x and want to estimate θ
 - We can update our estimate of θ based on observations x
- $\Pr(x)$ is the *evidence*, i.e., the probability of observing x , regardless of parameter θ

Computing the posterior distribution

- Given that we know (or we have guessed)

$$p(\theta | x) = \frac{p(x | \theta) p(\theta)}{p(x)} \propto p(x | \theta) p(\theta)$$

- In practice, it is usually not feasible to compute $p(\theta | x)$ in a closed form, as x and θ can be high-dimensional vectors
- Bayesian models in machine learning often use approximation techniques
- It can be difficult to determine a good prior $p(\theta)$; one possibility is to start with a non-informative prior and then update it using the formula

Computing the posterior distribution

- Given that we can find the new best estimate for parameter α , we can include α for θ from $\text{argmax}_{\theta} p(x | \theta) p(\theta)$

$$p(\theta | x, \alpha) = \frac{p(x | \theta) p(\theta)}{p(x)}$$

- In practice, as x and α are known, $p(\theta | x, \alpha)$ in a closed form,
- Bayesian approximation techniques
- It can be difficult to compute the posterior distribution; one possibility is to start with a non-informative prior and then update it using the formula

Note that the denominator is for normalisation, and we do not need it for finding the optimal θ

Bayesian inference: an example

Suppose we have **five coins**:

- Three unbiased: $p_1 = 0.5$
- One biased: $p_2 = 0.4$
- One biased: $p_3 = 0.6$

We **pick one coin at random** (unknown bias parameter $\theta \in \{0.5, 0.4, 0.6\}$).

Prior probabilities (since coins are chosen uniformly at random):

$$P(\theta = 0.5) = \frac{3}{5}, \quad P(\theta = 0.4) = \frac{1}{5}, \quad P(\theta = 0.6) = \frac{1}{5}.$$

Now suppose we **toss the coin 5 times** and observe:

$$x = \{1, 1, 1, 0, 1\}, \quad (4 \text{ heads, } 1 \text{ tail}).$$

Likelihood under parameter θ :

$$P(x \mid \theta) = \theta^4(1 - \theta)^1.$$

Posterior distribution (by Bayes' theorem):

$$P(\theta \mid x) = \frac{P(x \mid \theta) P(\theta)}{\sum_{\theta'} P(x \mid \theta') P(\theta')}.$$

Bayesian inference: an example

Priors (choose 1 of 5 coins at random):

$$P(\theta=0.5) = \frac{3}{5}, \quad P(\theta=0.4) = \frac{1}{5}, \quad P(\theta=0.6) = \frac{1}{5}.$$

Likelihood: $P(x \mid \theta) = \theta^k (1 - \theta)^{n-k} = \theta^4 (1 - \theta).$

Unnormalized posteriors:

- $w_{0.5} = 0.5^4 \cdot 0.5 \cdot \frac{3}{5} = 0.01875$
- $w_{0.4} = 0.4^4 \cdot 0.6 \cdot \frac{1}{5} = 0.003072$
- $w_{0.6} = 0.6^4 \cdot 0.4 \cdot \frac{1}{5} = 0.010368$

$$\text{Normalization } Z = 0.01875 + 0.003072 + 0.010368 = 0.03219.$$

Posterior (rounded):

$$\begin{aligned} P(\theta=0.5 \mid x) &\approx \frac{0.01875}{0.03219} = \mathbf{0.5825}, \\ P(\theta=0.4 \mid x) &\approx \frac{0.003072}{0.03219} = 0.0954, \\ P(\theta=0.6 \mid x) &\approx \frac{0.010368}{0.03219} = 0.3221. \end{aligned}$$

- **MAP estimate:** $\theta_{\text{MAP}} = 0.5.$
- **Posterior-predictive** $P(\text{Head next}): 0.5(0.5825) + 0.4(0.0954) + 0.6(0.3221) \approx \mathbf{0.5227}.$

Summary

- Understanding of probabilities is essential for understanding many concepts and techniques in non-symbolic AI
- This lecture is an introduction (or refresher) to the fundamentals of probability theory
 - Random variables, probability density functions
 - Expectation, variance, and standard deviation
 - Important probability distributions (uniform, Gaussian)
 - Joint and conditional probabilities
 - Bayes' theorem and its basic applications

Questions and discussion